6 **(a)** Define *acceleration*.

Acceleration is the rate of change of velocity [B1]

[1]

(b) State what is meant by a *field of force*.

A field of force due to a body's property (eg. charge, mass) is a region in space in which another body carrying that property experiences a force when it is placed in the field [B1] [1]

(c) Two parallel metal plates A and B are separated by a distance of 2.8 cm in a vacuum, as shown in Fig. 6.1.

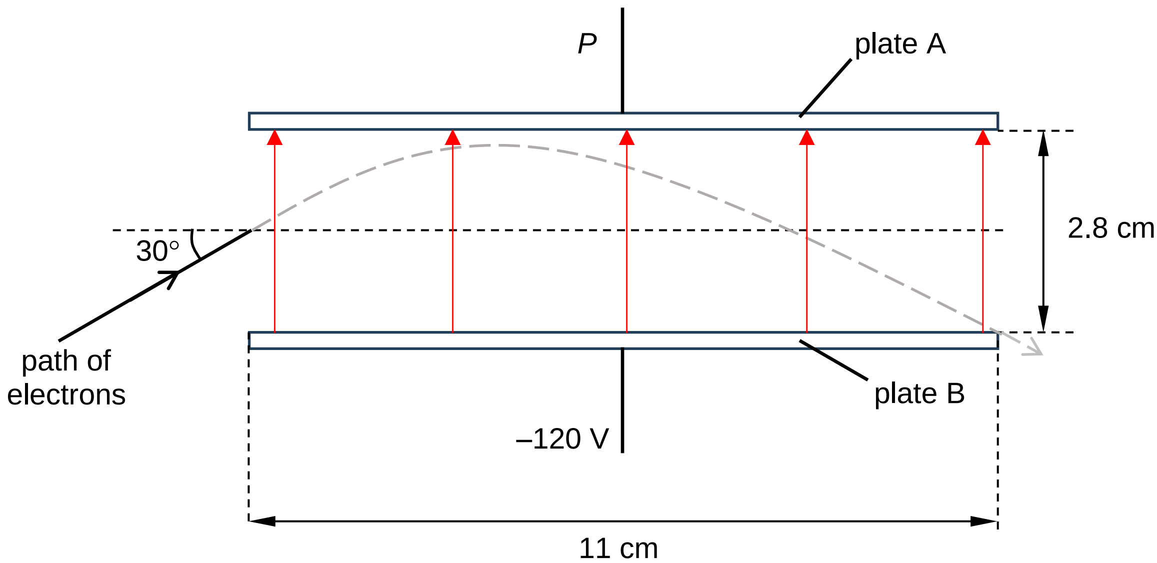

Fig. 6.1

The plates have length 11 cm. Plate A is at unknown potential *P* while plate B is at a potential of –120 V. The electric field may be assumed to be uniform between the plates and zero outside the plates.

An electron with kinetic energy 4.1×10^{-16} J enters the region midway between the plates. The initial direction of the electron is at an angle 30° above the horizontal. The electron charts out a parabolic path between the plates and exits just at the edge of plate B as shown in Fig. 6.1.

(i) Sketch on Fig. 6.1, lines to represent the electric field within the plates. [1]

(ii) For the electron between the metal plates,

1. show that the vertical component of velocity just as the electron enters the electric field is 1.5×10^7 m s^{-1},

$$\text{KE} = \frac{1}{2} m_e v^2 = 4.1 \times 10^{-16}$$

$$v = \sqrt{\frac{(2)(4.1 \times 10^{-16})}{9.11 \times 10^{-31}}} = 30.002 \times 10^6 \quad \text{B1}$$

$$v_y = v \sin 30°$$

$$= 1.50009 \times 10^7 \quad \text{B1}$$

$$= 1.5 \times 10^7 \text{ m s}^{-1} \quad \text{A0}$$

[2]

2. the time for the electron to travel a horizontal distance equal to the length of the plates,

$$s = u_x t$$
$$t = \frac{s}{u_x} = \frac{11\times10^{-2}}{30.0\times10^{6}\cos 30°}$$
$$= 4.23\times10^{-9}\ \text{s} \qquad \text{A1}$$

time = ... s [1]

3. calculate the acceleration of the electron.

Upwards taken as positive direction

$$s = u_y t + \frac{1}{2}at^2$$
$$-1.4\times10^{-2} = (1.5\times10^{7})(4.23\times10^{-9}) + \frac{1}{2}a(4.23\times10^{-9})^2$$
$$a = -8.66\times10^{15}\ \text{m s}^{-2}$$

acceleration = ... m s^{-2} [2]

(iii) Hence or otherwise, determine the potential *P* of plate A for the electrons to chart out the path shown in Fig. 6.1.

$$F = qE = q\frac{\Delta V}{d} = ma$$
$$\Delta V = \frac{mad}{q} = \frac{(9.11\times10^{-31})(8.66\times10^{15})(2.8\times10^{-2})}{1.6\times10^{-19}} \qquad \text{M1}$$
$$= 1.38\times10^{3}\ \text{V} \qquad \text{C1}$$

Plate A is at a lower potential:

$$\Delta V = 1380 = -120 - P$$
$$P = -1500\ \text{V} \qquad \text{A1}$$

P = ... V [3]

[Total: 11]

7 A square coil of side 5.0 cm and 50 turns is placed horizontally midway between the poles of a